Mandatory setup for real AI Engineer

Dev Environment Setup

🐍 STEP 1 — Python

🤔 WHY Python?
- Most popular language for AI/ML/GenAI in the industry
- 85%+ of data scientists & AI engineers use Python as their primary tool
- Simple, readable syntax = you focus on building, not fighting the language
- In simple words: You want to work in AI/ML → Python is non-negotiable ✅

📥 Install Python (need 3.11 or above):
🔗 https://www.python.org/downloads/

✅ After installing, verify in terminal:
python --version OR python3 --version

⚠️ Windows users: During install → CHECK the box "Add Python to PATH"

📹 Quick YouTube Tutorial (Python install in 5 mins):
https://www.youtube.com/watch?v=YYXdXT2l-Gg
(Programming with Mosh — trusted by millions of beginners)

💻 STEP 2 — Code Editor (VS Code)

🤔 WHY VS Code?
- Free & open-source (by Microsoft)
- Used in most companies and teams worldwide
- Not just a text editor — it's a full development environment
- Built-in terminal, Git integration, AI assistance (GitHub Copilot), debugging
- In simple words: It's where real AI engineer write real code 💻

📥 Download VS Code:
🔗 https://code.visualstudio.com/

After installing VS Code:
→ Open VS Code
→ Press Ctrl+Shift+X (Extensions)
→ Search "Python" by Microsoft → Install
→ Also search "Pylance" → Install (better IntelliSense/autocomplete)

📹 Quick YouTube Tutorial (VS Code + Python setup):
https://www.youtube.com/watch?v=W--_EOzdTHk
(Official VS Code Python setup)

📂 STEP 3 — Git & GitHub

🤔 WHY Git & GitHub?
- Git = version control. Think of it as "Ctrl+Z" for your entire project, forever.
- Every company in the world uses Git. No exception.
- GitHub = where your code lives on the cloud. Your coding portfolio.
- When you join a company → Day 1, they give you access to GitHub repo
- When you build projects → GitHub is your resume for technical roles
- In simple words: Without Git, one mistake can destroy your project. With Git, nothing is ever truly lost 🛡️

📥 Install Git:
🔗 https://git-scm.com/downloads

✅ Verify install:
```
git --version
```

👤 Set your Git profile (one-time setup in terminal):
```
git config --global user.name "Your Name"
git config --global user.email "your@email.com"
```

📝 Create GitHub Account (if you don't have one):
🔗 https://github.com/signup

📹 Quick YouTube Tutorial (Git + GitHub for Beginners):
https://www.youtube.com/watch?v=RGOj5yH7evk
(freeCodeCamp — most watched Git beginner video, ~1 hour, covers everything)

📌 QUICK CHECKLIST — Before Next Class

[ ] Python 3.11+ installed → python --version works
[ ] VS Code installed + Python extension added
[ ] Git installed → git --version works
[ ] GitHub account created
[ ] Git profile configured (name + email)

🚨 Stuck? No worries — drop a message in the group and we'll sort it out together!